

Pathfinder TN

An AI-powered career guidance platform built for Tunisian engineering students.

✨ What is Pathfinder TN?

Pathfinder TN is a free, open-source web application that helps Tunisian students discover their ideal engineering career path. Whether you are a high school graduate preparing for the national engineering entrance exam (*Concours National d'Accès aux Écoles d'Ingénieurs*), a CPGE student unsure which specialty to choose, or simply curious about engineering — Pathfinder guides you through the decision with **AI-powered recommendations** and **comprehensive field guides**.

🎯 The Problem We Solve

Every year, thousands of Tunisian students face the same dilemma:

- “Which engineering field should I choose?”
- “What do software engineers actually do?”
- “Where can I study robotics in Tunisia?”
- “How much do engineers earn here vs. abroad?”

Pathfinder TN answers all of these — in one place, in Darija-friendly English, with real Tunisian university data.

🚀 Features

AI Pathfinder Quiz

A 6-question personality & interest quiz powered by **Groq LLM (Llama 3)**. The AI analyzes your answers holistically and recommends the **2-3 best-fit engineering fields** with personalized explanations.

- **Smart fallback:** If the AI API is unavailable, a local scoring engine instantly provides results — the quiz never breaks.
- **Secure:** Users provide their own free Groq API key. No keys are stored on any server.

Explore 28 Engineering Fields

Deep-dive guides covering:

Section	Details
Curriculum	Year-by-year subjects (Prépa → Engineering cycle)
Universities	Exact schools in Tunisia —INSAT, ENIT, SUP'COM, ENIS, ENIM, EPT, and more
Careers	Real job titles with day-in-the-life schedules
Salaries	Tunisia vs. Europe vs. North America (entry & mid-level)

Table 1 – continued

Section	Details
Traits & Interests	“Is this right for me?” self-assessment
Future Tech	Technologies shaping each field (AI, quantum, green energy, etc.)

Roadmaps (Coming Soon)

Personalized learning plans based on your current level and goals.

Tech Today (Coming Soon)

A magazine covering the latest breakthroughs in engineering — AI, energy, robotics, and what’s next.

Covered Engineering Schools

Pathfinder TN includes data from Tunisia’s top engineering institutions:

- **INSAT** — Tunis (Génie Logiciel, Informatique Industrielle)
- **ENIT** — Tunis (Civil, Electrical, Mechanical, Industrial, ICT, Hydraulique)
- **SUP’COM** — Ariana (Telecom, Cybersecurity, Data Science)
- **ENSI** — Manouba (Computer Science, Intelligent Systems)
- **ENIS** — Sfax (Computer Science, Electrical, Mechanical, Civil)
- **ENIM** — Monastir (Mechanical, Energy, Textile, Industrial)
- **EPT** — La Marsa (Polytechnique, Advanced Techniques)
- **ENIG** — Gabès (Chemical, Civil, Petroleum)
- **ENISO** — Sousse (Mécatronique, Informatique Industrielle)
- **ENSIT** — Tunis (Topographie & Géomatique)
- **ENICarthage** — Ariana (Mécatronique, Electronics)
- **ISTMT** — Tunis (Biomedical)
- **INAT** — Tunis (Agronomic, Food Industry)
- **ESIAT** — Tunis (Food Industry)
- **And more...**

Tech Stack

Layer	Technology
Frontend	Vanilla HTML5, CSS3, JavaScript (ES6+)
Styling	Custom CSS with CSS Variables, Glassmorphism, Animations
AI Engine	Groq API (Llama 3 8B) —free tier available
Data	Static JSON (28 engineering fields)
Hosting	GitHub Pages / Netlify (100% static, zero server costs)

Quick Start

For Users

1. Visit the live site: https://YOUR_USERNAME.github.io/pathfinder-tn
2. Click “**Start Pathfinder**”
3. Get a free API key from console.groq.com/keys (takes 30 seconds)
4. Paste your key and take the 6-question quiz
5. Get your personalized AI recommendation!

For Developers

```
# Clone the repo
git clone https://github.com/YOUR_USERNAME/pathfinder-tn.git
cd pathfinder-tn

# Open with Live Server (VS Code extension) or any static server
# Do NOT double-click index.html — fetch() requires a server
```

No build step. No dependencies. No server to run.

Project Structure

```
pathfinder-tn/
├─ index.html      # Main HTML shell
├─ app.js          # All logic: router, quiz, AI call, rendering
├─ style.css       # Complete design system + animations
├─ data.json       # 28 engineering fields with full data
├─ favicon.png     # Site icon
└─ background.mp4  # Hero background video (optional)
```

Design Highlights

- **Dark theme** with gold & cyan accents — easy on the eyes for long reading sessions
 - **Glassmorphism cards** with backdrop blur and subtle borders
 - **3D flip cards** on the hero section (CSS `transform-style: preserve-3d`)
 - **Shimmering gradient title** animation on the hero
 - **Fully responsive** — works on mobile, tablet, and desktop
 - **Smooth scroll** and page transitions
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Privacy & Security

- **Your API key stays in YOUR browser** (localStorage / sessionStorage)
- **No backend server** — we never see your key or your quiz answers
- **No tracking** — no Google Analytics, no cookies, no data collection
- **Open source** — you can verify every line of code

👉 Contributing

We welcome contributions! Here are some ways to help:

- 🐛 **Report bugs** — open an issue with steps to reproduce
- 📝 **Update data** — salaries, new schools, or curriculum changes
- 🌍 **Add fields** — we currently cover 28; there are more!
- 🎨 **Improve design** — CSS animations, accessibility, dark mode toggle
- 🌐 **Translate** — add Arabic/French language support
- 🧠 **Enhance AI** — improve the prompt or scoring algorithm

Contribution Guidelines

1. Fork the repository
2. Create a feature branch: `git checkout -b feature/your-feature`
3. Commit your changes: `git commit -m "Add: your feature"`
4. Push to the branch: `git push origin feature/your-feature`
5. Open a Pull Request

📖 License

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🙏 Acknowledgments

- **Groq** for providing fast, free LLM inference
- **Tunisia's engineering schools** for shaping generations of engineers
- **Every student** who ever asked "*Which field should I choose?*" — this is for you

Contact

Have questions or feedback?

- Open an [Issue](#)
- Or reach out on [LinkedIn / Twitter / Email]

 Built with pride for Tunisian engineering students  Aslema. Marhbe. Welcome to your future.